NUR-E-JANNAT ANIKA

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EDUCATION

Colby College May 2028
Bachelor of Arts in Computer Science, Neuroscience *Waterville, Maine*

- **Relevant Coursework:** Computational Thinking with Computer Vision; Data Structures & Algorithms; Data Analysis and Visualization; Applied Biomedical Genomics; Series & Multivariable Calculus; Linear Algebra; Vector Calculus; Cellular Basis of Life; Evolutionary Biology; Neurobiology

SKILLS

- **Programming:** Python (NumPy, Pandas, Matplotlib, Seaborn, SciPy, Scikit-learn, Scikit-bio, TensorFlow, OpenCV, BioPython); Java; C; R
- **Genomics & Bioinformatics:** Hifiasm, YaHS, Juicebox, BUSCO, Merqury, BRAKER3, RepeatModeler/RepeatMasker, OrthoFinder, CAFE5, minimap2, samtools, DIAMOND, 16S amplicon analysis, ANCOM
- **ML & Data Science:** Random Forest, SVM, XGBoost, Siamese Neural Networks, PCoA, PERMANOVA; HPC (SLURM), Git, Conda

RELEVANT EXPERIENCE (TECH & RESEARCH)

Angelini Lab, Colby College | Undergraduate Researcher Waterville, ME | May 2026 – Present
Advisor: Dr. David R. Angelini, Dept. of Biology

- **Chromosome-Level Genome Assembly:** Producing the first reference genome for the soapberry bug J. haematoloma (Hemiptera). Wrote Bash and Python scripts on HPC (SLURM) to run certain software— de novo assembly (hifiasm), scaffolding (YaHS), contact map curation (Juicebox), quality assessment (BUSCO, Merqury QV50+ target). Writing further scripts for next steps (in progress).
- **Gene Family Evolution:** Wrote an end-to-end computational pipeline (Bash/SLURM + Python + R) for CAFE5 analysis of gene gain/loss across 11 Bombus species + outgroup: automated NCBI protein download, isoform filtering (BioPython), OrthoFinder clustering, ultrametric tree generation (R/phytools), and CAFE5 λ estimation — all as reproducible SLURM batch jobs. Currently writing abstract.
- **Insect Gut Microbiome:** Wrote a complete analysis pipeline from scratch in Python (no QIIME 2) for 16S rRNA V4 amplicon data (204 samples, 2,680 ASVs) from J. haematoloma: implemented rarefaction, alpha diversity (Shannon, Simpson, Pielou's evenness), Bray-Curtis beta diversity, PCoA ordination, PERMANOVA, differential abundance testing with FDR correction, and a custom ANCOM implementation. Found host plant and geography significantly structured microbiome composition. Currently writing paper.

Colby College (Prof. Eric Aaron) | Undergraduate Research Assistant Waterville, ME | Jun – Nov 2025

- Enhanced an Embodied Computational Evolution (ECE) model simulating evolutionary dynamics in digital organisms, improving simulation stability across 1,000+ runs and reducing average runtime by 33%.
- Discovered and resolved 3 critical bugs; presented findings at CUSRR (100+ attendees) and earned co-authorship on a forthcoming research publication.

Early Detection of Parkinson's Disease | Independent Research Feb 2023 – Aug 2023

- Analyzed 195 patient voice recordings; benchmarked 5+ supervised ML models, achieving 97% accuracy with Random Forest.
- Accepted for poster presentation at **AAN 2024 Annual Meeting** and publication in *Journal of Advanced Zoology*.

PROJECTS

AI Travel Storybook & Memory Engine | Hackathon (Google Gemini Track) March 2026

- Built the backend and AI integration for a full-stack app (Node.js/Express + React/Vite) that transforms travel photos into literary storybooks with audio narration, using Google Gemini 2.5 Flash with Search grounding and ElevenLabs TTS.
- Engineered the story generation pipeline: photo-to-base64 conversion, per-place Google Search grounding for real-world context, multi-photo scene analysis, and structured JSON output producing chapters, timelines, and reflections. Working on it to make it a full product

FaceGuard: Real-Time Face Verification | CS166 Nov 2024

- Built a real-time face verification system for secure authentication that compares live camera input against stored facial profiles to confirm identity. Designed and trained a Siamese Neural Network (OpenCV + TensorFlow) using contrastive loss to learn facial similarity embeddings, achieving ~92% verification accuracy across 500 test images and rejecting unauthorized faces in 80%+ of cases across varying lighting conditions.

FELLOWSHIPS

Equitech Futures Institute — EFI Scholar (Fully Funded) Jun - Aug 2026

- Completed 8-week intensive classes covering AI system architecture for societal challenges (feasibility assessment, data collection, model selection, failure-mode analysis), data storytelling and science communication, and governance of frontier AI.
- Completed capstone projects on AI policy design, a blueprint for evaluating top-to-bottom of an AI system, and personal storytelling

LEADERSHIP

Halloran Lab for Entrepreneurship | *Student Advisory Board Member* *Colby College* | Nov 2025 – Present

- Selected through a competitive process as 1 of 12 founding student advisors to shape innovation and entrepreneurship programming.
- Conducted qualitative user research through structured student interviews to identify participation gaps and access barriers among underrepresented student populations.
- Advising senior leadership on pilot initiatives, program design, and outreach strategy, contributing student-driven insights that influence programming, impacting 400+ students.

DavisAI Institute | *Strategic Communications Lead* *Colby College* | Sep 2025 – Present

- Designing and executing the organization's communications strategy in collaboration with the technical team, increasing average event attendance by 40%.
- Managing the official newsletter, weekly student meetings, and 4+ communication channels; coordinating with student presenters, faculty, and external speakers to deliver interdisciplinary programming and expand participation beyond CS majors.